Name:_____

TOPIC TEST 1

Chapter 3 Applications of derivatives

- Time allowed: 70 minutes
- CAS calculators may be used.
- Part A: Five multiple-choice questions (5 marks).
- Part B: Six short-answer questions (34 marks).
- Part C: Two analysis questions (16 marks).

Part A

Multiple-choice questions

Five multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Given that the equation of a curve is $y = 3x^3 + 4x^2 + 5$, find the approximate increase in y as x increases from 2 to 2.03.

A 45

B 46.58

C 52

D 1.56

E 1.58

[1 mark]

2 The second derivative of $y = 4x\cos(x)$ is:

A $4\cos(x) - 4x\sin(x)$

B $-4x\cos(x) + 4\sin(x)$

C $-4x\cos(x) - 8\sin(x)$

D $-4x\cos(x) - 4\sin(x)$

E $-4x\cos(x) + 8\sin(x)$

[1 mark]

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3 For what values of x is the curve $y = 2x^3 + 12x^2 - 18x - 5$ concave upwards?

A $-3 < x < -1$

B $x > -2$

C $x > 2$

D $1 < x < 3$

E $x < -2$

[1 mark]

4 The graph of the function $y = x^3 - 6x^2 - 36x + 9$ has the following stationary points:

A Minimum at $(2, -79)$ and maximum at $(-6, 207)$

B Minimum at $(2, -49)$ and maximum at $(-6, 207)$

C Minimum at $(-6, -207)$ and maximum at $(2, 49)$

D Minimum at $(6, -207)$ and maximum at $(-2, 79)$

E Minimum at $(6, -207)$ and maximum at $(-2, 49)$

[1 mark]

5 The product of two positive numbers is 72. Find the numbers such that the sum of twice one number and 4 times the other is a minimum.

A 6 and 12

B 4 and 18

C 8 and 9

D 2 and 36

E 3 and 24

[1 mark]

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Part B

Short-answer questions

Six short-answer questions
34 marks
Show your working where appropriate.

1 Given $k = -n^3 + 7n^2 + 8n + 11$, find $\frac{d^2k}{dn^2}$ when $n = -2$.

[3 marks]

2 The displacement of an object from a particular point is given by $s = 2t^3 + 9t - 8$, where s is in metres and t is in seconds.

a Find the velocity at 2 s.

b Find when the velocity is 15 m/s.

c Find the acceleration at 2 s.

d Find the acceleration when the velocity is 15 m/s.

[9 marks]

3 Sketch a curve where $f''(x) > 0$ for $x < 0$, $f''(x) < 0$ for $x > 0$ and $f'(x) < 0$ for all x.

[3 marks]

4 The average age, A, of the population of a shire was studied over time, and at the start the average age was 38.

a During the study, $\frac{dA}{dt} < 0$. What does this say about the age of the population during the study?

b If at the same time, $\frac{d^2A}{dt^2} < 0$, what can you say about the population rate?

c Sketch the graph of the population A against t.

[5 marks]

5 A baseball player strikes the ball so that its equation of motion is given by $h = 2x - 16 - 0.05x^2$, where h is the height (in metres) reached by the ball and x is the horizontal distance (in metres) travelled by the ball. What is the greatest height reached by the ball?

[6 marks]

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6 A soft drink manufacturer produces cylindrical aluminum cans, each with a volume of 500 mL (i.e. 500 cm^3). What is the radius of the can if the least amount of material is to be used in its construction? Give your answer correct to 2 significant figures.

[8 marks]

Part C

Analysis questions

Two analysis questions
16 marks
Show your working where appropriate.

1 The curve $y = x^3 - ax^2 - 27x + 1$ has a point of inflection where $x = 4$.

a Find the value of a.

b Hence, or otherwise, determine the stationary points on the curve and their nature.

[9 marks]

2 A slow-moving river is one kilometre wide. Tom wants to return to his camp on the opposite side of the river. He can swim at 2 km/h and walk at 3 km/h. Tom must first swim across the river to any point on the opposite bank then, from there, walk to his camp, which is one km from the point directly across the river from where he started his swim. What route will take the least amount of time? How long would this route take?

[7 marks]

[Total marks: 55]

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Answers

Part A – Multiple-choice questions [1 mark each]

1 $y = 3x^3 + 4x^2 + 5$

$y' = 9x^2 + 8x$

When $x = 2$

$y' = 52$

$\delta y = 52 \times 0.03$

$= 1.56$

$\therefore$ **D**

2 $y = 4x\cos(x)$

$y' = 4\cos(x) - 4x\sin(x)$

$y'' = -4\sin(x) - 4\sin(x) - 4x\cos(x)$

$= -8\sin(x) - 4x\cos(x)$

$\therefore$ **C**

3 $y = 2x^3 + 12x^2 - 18x - 5$

$y' = 6x^2 + 24x - 18$

$y'' = 12x + 24$

concave upwards when $y'' > 0$

$12x + 24 > 0$

$12x > -24$

$x > -2$

$\therefore$ **B**

4 $y = x^3 - 6x^2 - 36x + 9$

$y' = 3x^2 - 12x - 36$

For stationary points, $y' = 0$

$3x^2 - 12x - 36 = 0$

$x^2 - 4x - 12 = 0$

$(x - 6)(x + 2) = 0$

$x = -2, 6$

$y'' = 6x - 12$

When $x = -2$

$y'' = 6 \times -2 - 12 < 0$

Maximum at $(-2, 49)$

When $x = 6$

$y'' = 6 \times 6 - 12 > 0$

Minimum at $(6, -207)$

$\therefore$ **E**

5 Let the two numbers be x and y.

Then $xy = 72$ and the sum $S = 2x + 4y$

$$y = \frac{72}{x}$$

Substitute into $S = 2x + 4y$

$$S = 2x + 4\left(\frac{72}{x}\right)$$

$$S = 2x + \frac{288}{x}$$

$$\frac{dS}{dx} = 2 - \frac{288}{x^2}$$

Stationary point when $\frac{dS}{dx} = 0$,

$$2 - \frac{288}{x^2} = 0$$

$$2x^2 = 288$$

$x^2 = 144$, since x is positive

$x = 12$

$y = \frac{72}{12}$

$y = 6$

$\therefore$ **A**

Part B – Short-answer questions

1 $\frac{dk}{dn} = -3n^2 + 14n + 8$ **[1 mark]**

$\frac{d^2k}{dn^2} = -6n + 14$ **[1 mark]**

When $n = -2$

$\frac{d^2k}{dn^2} = -6(-2) + 14$ **[1 mark]**

$= 26$

2 a $s = 2t^3 + 9t - 8$

$\frac{ds}{dt} = 6t^2 + 9$ **[1 mark]**

When $t = 2$

$\frac{ds}{dt} = 6(2)^2 + 9$ **[1 mark]**

$= 33$ m/s

b 1 s

$\frac{ds}{dt} = 6t^2 + 9$

When $\frac{ds}{dt} = 15$

$6t^2 + 9 = 15$ **[1 mark]**

$6t^2 = 6$

$t^2 = 1$

$t = \pm 1$ **[1 mark]**

but $t \geq 0$, so $t = 1$ **[1 mark]**

c $\frac{d^2s}{dt^2} = 12t$ **[1 mark]**

When $t = 2$

$\frac{d^2s}{dt^2} = 12(2)$ **[1 mark]**

$= 24 \text{ m/s}^2$

d 12 m/s^2

velocity = 15 when $t = 1$ **[1 mark]**

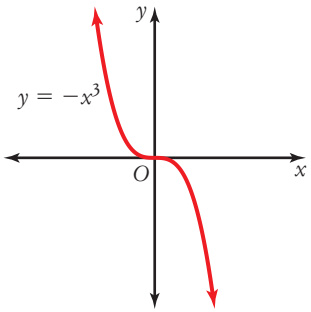
$\frac{d^2s}{dt^2} = 12(1)$

$= 12$ **[1 mark]**

3 Concave upwards for $x < 0$ **[1 mark]**

Concave downwards for $x > 0$ **[1 mark]**

Decreasing curve **[1 mark]**



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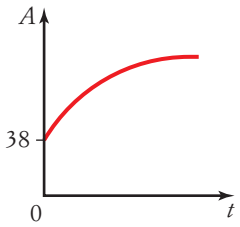
4 a The average age of the population is increasing.

[1 mark]

b The rate at which the age of the population is increasing is slowing down.

[1 mark]

c 1 mark for concave downwards for x
1 mark for increasing curve
1 mark for y-intercept of 38



5 $h = 2x - 16 - 0.05x^2$

$h' = 2 - 0.1x$ **[1 mark]**

Stationary point when $h' = 0$ **[1 mark]**

$2 - 0.1x = 0$

$x = 20$ **[1 mark]**

$h'' = -0.1 < 0$ maximum **[1 mark]**

When $x = 20$

$h = 2 \times 20 - 16 - 0.05(20)^2$ **[1 mark]**

$= 4$ m **[1 mark]**

6 Volume $= \pi r^2 h$

$500 = \pi r^2 h$ **[1 mark]**

$h = \dfrac{500}{\pi r^2}$ **[1 mark]**

Surface area $= 2\pi r^2 + 2\pi rh$

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$A = 2\pi r^2 + 2\pi r \times \frac{500}{\pi r^2}$ **[1 mark]**

$= 2\pi r^2 + \frac{1000}{r}$

$\frac{dA}{dr} = 4\pi r - \frac{1000}{r^2}$ **[1 mark]**

$= 0$ when $4\pi r - \frac{1000}{r^2} = 0$ **[1 mark]**

$4\pi r = \frac{1000}{r^2}$

$r^3 = \frac{1000}{4\pi}$

$r = \sqrt[3]{\frac{1000}{4\pi}}$

$r = 4.3$ correct to 2 sig. fig. **[1 mark]**

$\frac{d^2A}{dr^2} = 4\pi + \frac{1000}{r^3}$ **[1 mark]**

> 0 for all $r \geq 0 \therefore$ minimum **[1 mark]**

i.e. radius of can is 4.3 cm

Part C – Analysis questions

1 a 12

$y = x^3 - ax^2 - 27x + 1$

$y' = 3x^2 - 2ax - 27$ **[1 mark]**

$y'' = 6x - 2a$ **[1 mark]**

$y'' = 0$ when $6x = 2a$ **[1 mark]**

$a = 3x$

When $x = 4$, $a = 12$ **[1 mark]**

b $y' = 3x^2 - 24x - 27$ **[1 mark]**

$y' = 0$ when $3x^2 - 24x - 27 = 0$ **[1 mark]**

$x^2 - 8x - 9 = 0$

$(x - 9)(x + 1) = 0$

$x = 9, -1$ **[1 mark]**

When $x = 9$

$y'' = 6 \times 9 - 24 > 0$

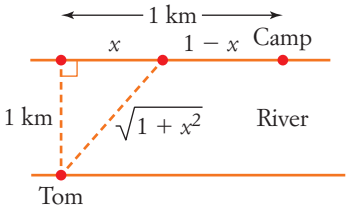
i.e. minimum at $(9, -485)$ **[1 mark]**

When $x = -1$

$y'' = 6 \times (-1) - 24 < 0$

i.e. maximum at $(-1, 15)$ **[1 mark]**

2 Swim to a point approximately 0.89 km along the river towards his camp and then walk approximately 0.11 km to his camp. This will take approximately 42 minutes 22 seconds.



Swim: 2 km/h, Walk: 3 km/h

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

Time = Swim time + Walk time **[1 mark]**

$$T = \frac{\sqrt{1+x^2}}{2} + \frac{1-x}{3}$$ **[1 mark]**

$$\frac{dT}{dx} = \frac{x}{2\sqrt{1+x^2}} - \frac{1}{3}$$ **[1 mark]**

$$= 0 \quad \text{when} \quad \frac{x}{2\sqrt{1+x^2}} - \frac{1}{3} = 0$$ **[1 mark]**

$$\frac{x}{2\sqrt{1+x^2}} = \frac{1}{3}$$

$$3x = 2\sqrt{1+x^2}$$

$$9x^2 = 4\left(1+x^2\right)$$

$$9x^2 = 4 + 4x^2$$

$$5x^2 = 4$$

$$x^2 = \frac{4}{5}$$ **[1 mark]**

$$x = \frac{2}{\sqrt{5}} \approx 0.89 \quad \text{since } 0 \leq x \leq 1$$ **[1 mark]**

Substitute into T to find $T \approx 0.706$ hours $\approx$ 42 minutes 22 seconds **[1 mark]**

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